

Assignment for Week 4

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01/31/2017

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Assignment

1) 19.2.3 (a) (p. 405)

$$\int (2x^3 + 3x - 1)dx = \frac{1}{2}x^4 + \frac{3}{2}x^2 - x + c,$$

where $c \in \mathbb{R}$ is a constant.

2) 19.2.6 (p. 406)

$$\begin{aligned}\int_0^1 \frac{2}{x+3} dx &= 2 (\ln |x+3|) \Big|_0^1 \\ &= 2 \ln 4 - 2 \ln 3 \\ &= 2 \ln \frac{4}{3}.\end{aligned}$$

If the range of integration is $[-5, -4]$, then

$$\begin{aligned}\int_{-5}^{-4} \frac{2}{x+3} dx &= 2 (\ln |x+3|) \Big|_{-5}^{-4} \\ &= 2 \ln 1 - 2 \ln 2 \\ &= -2 \ln 2.\end{aligned}$$

Since $\frac{2}{x+3}$ is defined on $\mathbb{R} \setminus \{-3\}$, the integral is not well-defined if the range of integration is $[-4, -2]$.

3) 11.1.4 (c, e, f) (p. 207)

(c) Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \text{ and } \mathbf{v}_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}.$$

Since $\mathbf{v}_1 - 2\mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$, the sets of vectors are linearly dependent.

(d) Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 2 \\ -2 \\ -1 \end{bmatrix}, \text{ and } \mathbf{v}_3 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}.$$

Suppose $a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3 = \mathbf{0}$, where $a, b, c \in \mathbb{R}$ are constants. Then we have

$$\begin{cases} a + 2b + 2c = 0 \\ 2a - 2b + c = 0 \\ a - b + 3c = 0 \end{cases}$$

The only solution to the above set of equations is $a = b = c = 0$. Therefore, the sets of vectors are not linearly dependent.

(e) Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 2 \\ -2 \\ 4 \end{bmatrix}, \text{ and } \mathbf{v}_3 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}.$$

Since $2\mathbf{v}_1 + \mathbf{v}_2 - 2\mathbf{v}_3 = \mathbf{0}$, the sets of vectors are linearly dependent.

4) 11.2.4 (p. 213)

The equation in matrix form is:

$$\begin{bmatrix} 1 & -3 & 1 \\ 2 & -4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \mathbf{0}$$

Let

$$A = \begin{bmatrix} 1 & -3 & 1 \\ 2 & -4 & 4 \end{bmatrix}.$$

When $\mathbf{x} = \mathbf{0}$, we apparently have $A\mathbf{x} = \mathbf{0}$, and so $\mathbf{x} = \mathbf{0}$ is a solution to the above equation.

We can easily verify that when

$$\mathbf{x} = \mathbf{c} = \begin{bmatrix} -4 \\ -1 \\ 1 \end{bmatrix} = \mathbf{0},$$

We also have $A\mathbf{x} = A\mathbf{c} = \mathbf{0}$, and so $\mathbf{x} = \mathbf{c}$ is a solution to the above equation.

Since for all $\lambda \in \mathbb{R}$, we have $A(\lambda\mathbf{c}) = \lambda(A\mathbf{c}) = \mathbf{0}$, then for all $\lambda \in \mathbb{R}$, $\lambda\mathbf{x} = \mathbf{c}$ is a solution to the above equation.

5) 11.2.6 (p. 213)

The mapping in 3.4.1 is $h(x, y) = (x, -y)$. Suppose the 2×2 matrix A can represent the mapping, i.e. $A(x, y) = (x, -y) \forall x, y \in \mathbb{R}$. Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Then

$$\begin{cases} ax + by = x \\ cx + dy = y \end{cases}, \text{ or } \begin{cases} (a-1)x + by = 0 \\ cx + (d+1)y = 0 \end{cases} \quad \forall x, y \in \mathbb{R}.$$

Since x, y can be any value, we let $x = 0$, and so $b = 0, d = -1$. Let $y = 0$, we have $a = 1, c = 0$. Therefore, the 2×2 matrix A representing the mapping is

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

6) 11.3.1 (p. 217)

$$AB = \begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 1 & -2 \\ 5 & -2 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 2 & 7 \\ 2 & 0 & -2 \\ -3 & -2 & -9 \\ 1 & -2 & -13 \end{bmatrix}.$$

$$AC = \begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 1 & -2 \\ 5 & -2 \end{bmatrix} \begin{bmatrix} 4 & -4 & 3 & 1 \\ 1 & 0 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 6 & -4 & 1 & 1 \\ 8 & -8 & 6 & 2 \\ 2 & -4 & 5 & 1 \\ 18 & -20 & 17 & 5 \end{bmatrix}.$$

$$CA = \begin{bmatrix} 4 & -4 & 3 & 1 \\ 1 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 1 & -2 \\ 5 & -2 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}.$$

7) Which of the following datasets are tidy?

Tidy data is data that has exactly one observation per row, with columns for each variable. In other words, each row characterise an observed outcome, and each column is a characteristic of that observed outcome.

```
dta1<- data.frame(year = c(2000:2003,2000:2003),
  CountryA = c(rep("USA",4),rep("CAN",4) ),
  CountryB=c(rep("Russia",4),rep("Russia",4) ),
  Agreement=c(0,1,1,0,0,1,0,0))
```

dta1 is tidy, since there is one observation per row, here agreement between two countries in a certain year, and each column is a characteristic of the agreement.

```
dta2<- data.frame(year = c(2000:2003,2000:2003),
  Dyad = c(rep("USA-RUS",4),rep("USA-RUS",4) ),
  ConflictType=c(rep("War",4),rep("Minor MID",4) ),
  ConflictLevel=c(0,3,1,0,0,1,0,0))
```

dta2 is tidy, since there is one observation per row, here conflicts between two countries in a certain year, and each column is a characteristic of the conflict.

```
dta3<- data.frame(Dyad = c("USA-RUS", "CAN-RUS"),
  War2002=c(0,2),
  War2003=c(0,2),
  War2004=c(0,2),
  MinorMID2002=c(0,2),
  MinorMID2003=c(0,2),
  MinorMID2004=c(0,2))
```

dta3 is untidy, with wars and levels of conflict along the top, and variable names in each of the rows.

8) Use spread, gather and separate from the tidy package to fix the untidy data.frames from question 7.

```
dta3 %>% select(`Dyad`, `War2002`:`War2004`, `MinorMID2002`:`MinorMID2004`) %>% gather(key = ConflictType
```

```
##      Dyad ConflictType ConflictLevel
## 1  USA-RUS      War2002              0
## 2  CAN-RUS      War2002              2
## 3  USA-RUS      War2003              0
## 4  CAN-RUS      War2003              2
## 5  USA-RUS      War2004              0
## 6  CAN-RUS      War2004              2
## 7  USA-RUS MinorMID2002              0
## 8  CAN-RUS MinorMID2002              2
## 9  USA-RUS MinorMID2003              0
## 10 CAN-RUS MinorMID2003              2
## 11 USA-RUS MinorMID2004              0
## 12 CAN-RUS MinorMID2004              2
```

9) Correct the common mistakes in the following code snippets

a) Make the following TRUE

```
mean(c(3000, 2, 1, 0, 4)) == 601.4
```

```
## [1] TRUE
```

b) Make the following into a vector TRUE TRUE TRUE

```
EuclidDistancetoKabul <- function(x) {sqrt((-64 - x[1]) ^ 2 + (16 - x[2]) ^ 2)}  
x <- EuclidDistancetoKabul(rbind(c(48, 4), c(12, 3), c(80, 13)))  
x > c(20, 20, 20)
```

```
## [1] TRUE TRUE TRUE
```

c) Make the following code install the 'car' package.

```
install.packages("car", repos="http://cran.rstudio.com/", dependencies = TRUE)
```

```
## Installing package into 'C:/Users/Grace/Documents/R/win-library/3.3'
```

```
## (as 'lib' is unspecified)
```

```
## package 'car' successfully unpacked and MD5 sums checked
```

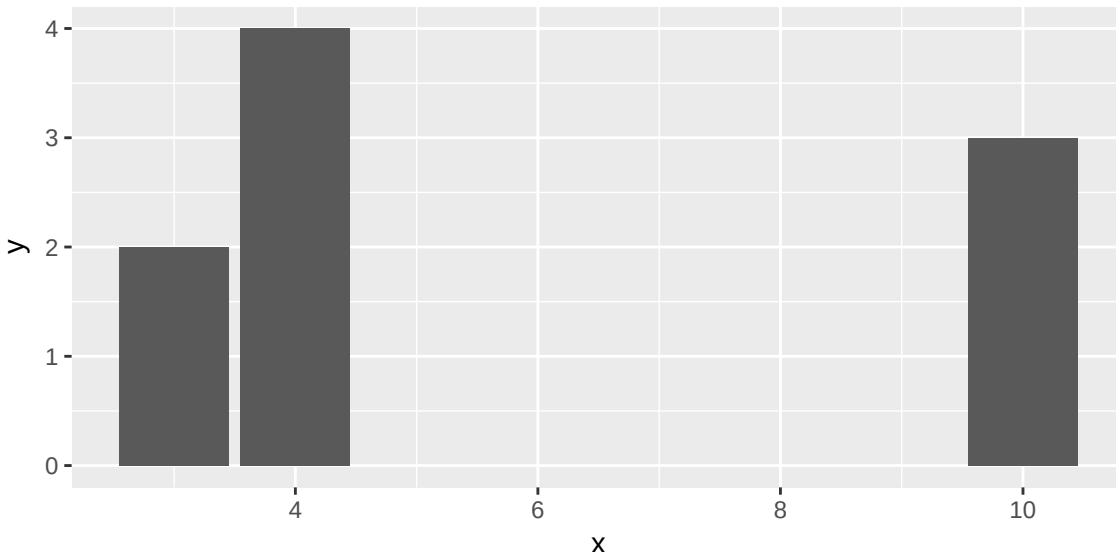
```
##
```

```
## The downloaded binary packages are in
```

```
## C:\Users\Grace\AppData\Local\Temp\Rtmpkf0fSE\downloaded_packages
```

d) Correct the following plot command:

```
x <- c(3, 10, 4)  
y <- c(2, 3, 4)  
df <- data.frame(x, y)  
ggplot(df, aes(x, y)) + geom_bar(stat="identity")
```



e) Make the following code report TRUE

```
allthetypes <- data.frame(x = c("Iowa"), stringsAsFactors = F)  
is.character(allthetypes[1,1])
```

```
## [1] TRUE
```

- 10) How many shootings occur in the state with the most active VA hospitals? The least active?
 Here I choose the number of “Total Service Users” to represent VA hospital activity.
 In this case, the state with the most active VA hospital is Florida, where 122 shootings occurred.
 The state with the least active VA hospital is Wyoming, where 8 shootings occurred.

```
# Read a dataset of all police shootings in 2015:
shootings <- read.csv("https://raw.githubusercontent.com/washingtonpost/data-police-shootings/master/fatality_shootings.csv")

library(jsonlite)
# Read data from the government on Veterans Administration services:
VAest <- fromJSON(paste0("https://raw.githubusercontent.com/vacobrydsk/",
                        "VHA-Files/master/NEPEC_AnnualDataSheet_MH_FY15.json"), flatten = T)

# Make "shooting" data tidy, and rename columns:
VAtidy <- VAest %>% select(-ValueType, -Category) %>% spread(Item, Value)
fixednames <- c("Station", "ADC_all_beds", "ADC_inpatient", "ADC_residential",
"ADC_psych_encounters", "ADC_psych_unique", "Avestops_MH",
"Avestops_Psychiatry", "Avestops_Substance", "Setting_Inpatient",
"Setting_Outpatient", "Setting_Resident", "N_serviceusers",
"N_serviceusers_confirmed", "N_serviceusers_possible", "Pct_confirmed",
"Pct_possible", "NMH_Prop_community_confirmed", "NMH_Prop_inpatient_confirmed",
"NMH_Prop_outpatient_confirmed", "NMH_Prop_community_possible",
"NMH_Prop_inpatient_possible", "NMH_Prop_outpatient_possible",
"Prop_Vets_confirmed_any", "Prop_Vets_confirmed_outpatient",
"Prop_Vets_confirmed_inpatient", "Prop_Vets_confirmed_resident",
"Prop_Vets_possible_any", "Prop_Vets_possible_outpatient",
"Prop_Vets_possible_inpatient", "Prop_Vets_possible_resident",
"testing_encounters", "testing_uniques", "ServiceUserswithMH",
"Total_NHM_stays_community", "Total_NHM_stays_inpatient",
"Total_NHM_stays_outpatient", "Total_Service_Users", "Unique_Vets_inpatient",
"Unique_Vets_outpatient", "Unique_Vets_resident")
names(VAtidy) <- fixednames

# Change data type from string to numeric:
library(stringr)
stringtonumber <- function(sortanumber) {as.numeric(str_replace_all(as.character(sortanumber), "%|,", ""))}
VAtidy2 <- VAtidy %>% mutate_all(stringtonumber)

## Warning in `__dplyr_colwise_fun`(c("358", "402", "405", "436", "437",
## "438", : NAs introduced by coercion

# Read data of the locations of each VA station, and rename its column "facility_id" to "Station":
VAFacilities <- fromJSON("https://raw.githubusercontent.com/departement-of-veterans-affairs/VHA-Facilities/master/VHA-Facilities.json")
VAFacdata <- VAFacilities$VAFacilityData %>% mutate(Sttn = facility_id)

# Connecting VA hospital data with VA station locations:
library(dplyr)
VAtidy2 <- VAtidy2 %>% select(Sttn= Station, `Total Service Users` = `Total_Service_Users`)
VAFacdata2 <- merge(VAFacdata, VAtidy2, by="Sttn")
shootingbystate <- shootings %>% group_by(state) %>% summarize (nshot= length(race))
facilitywshooting <- merge (shootingbystate, VAFacdata2, by="state")

# To find the state with the most active VA hospitals:
facilitywshooting2 <- facilitywshooting[order(-facilitywshooting$`Total Service Users`), ]
facilitywshooting2[1:3,1:2, drop= TRUE]
```

```
##      state nshot
## 60     FL   122
## 67     FL   122
## 75     GA    60
```

```
# To find the state with the least active VA hospitals:
```

```
facilitywshooting3 <- facilitywshooting[order(facilitywshooting$`Total Service Users`), ]
facilitywshooting3[1:4,1:2, drop= TRUE]
```

```
##      state nshot
## 323     WY     8
## 326     WY     8
## 328     WY     8
## 329     WY     8
```

- 11) Plot the following function for $x \in [-4, 4]$ by writing a custom function(). Use `integrate()` on that function to plot the area under the curve of that function from $-\infty$ to x , such that $x \in [-4, 4]$.

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

I am approximating the area under the curve from $-\infty$ to 0 by plotting $[-5, 0]$ under the curve. The upper limit of integration can be changed by adjusting the second argument of `B`.

```
A <- function(x) {
y <- (1/sqrt(2*pi))*(exp((-x^2)/2))
return(y)
}
B <- function(llim,ulim) {
x <- seq(-100,100,0.01)
y <- A(x)
x_lim <- seq(llim,ulim,0.05)
plot(x,y, cex= 0.1, xlim = c(-5,5))
for (i in 1:length(x)){
coord.x <- c(x_lim[i],x_lim[i], x_lim[i+1],x_lim[i+1])
coord.y <- c(0,A(x_lim[i]), A(x_lim[i+1]),0)
polygon(coord.x,coord.y,col='gray', border=NA)
} }
B(-5,0)
```

